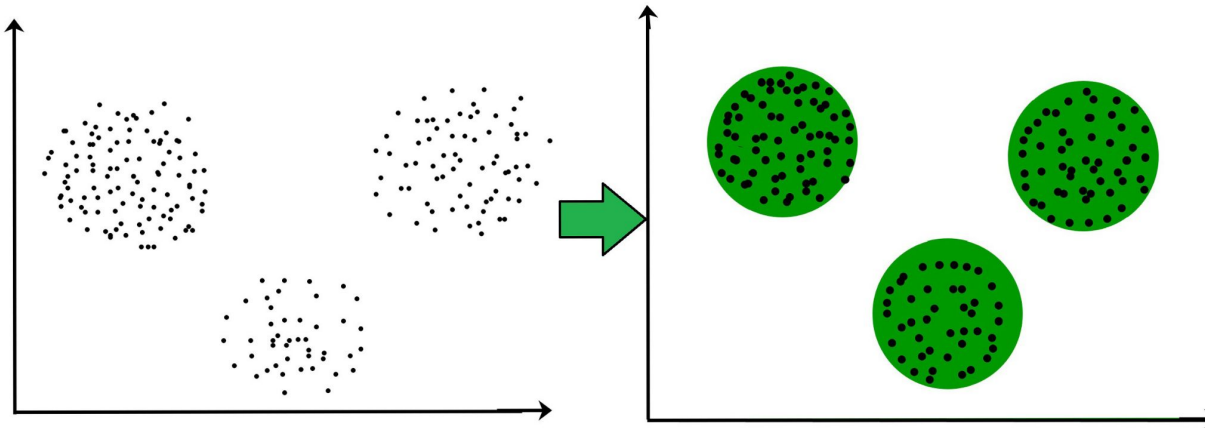


Cluster analysis

k-means

K-Means

- K-Means is an unsupervised learning algorithm that partitions n observations into **k clusters**
- Each observation belongs to the cluster with the nearest mean (centroid)



How it works

Step-by-step process:

Step 1: Choose the number of clusters (k)

Step 2: Randomly initialize k centroids

Step 3: Assign each data point to the nearest centroid

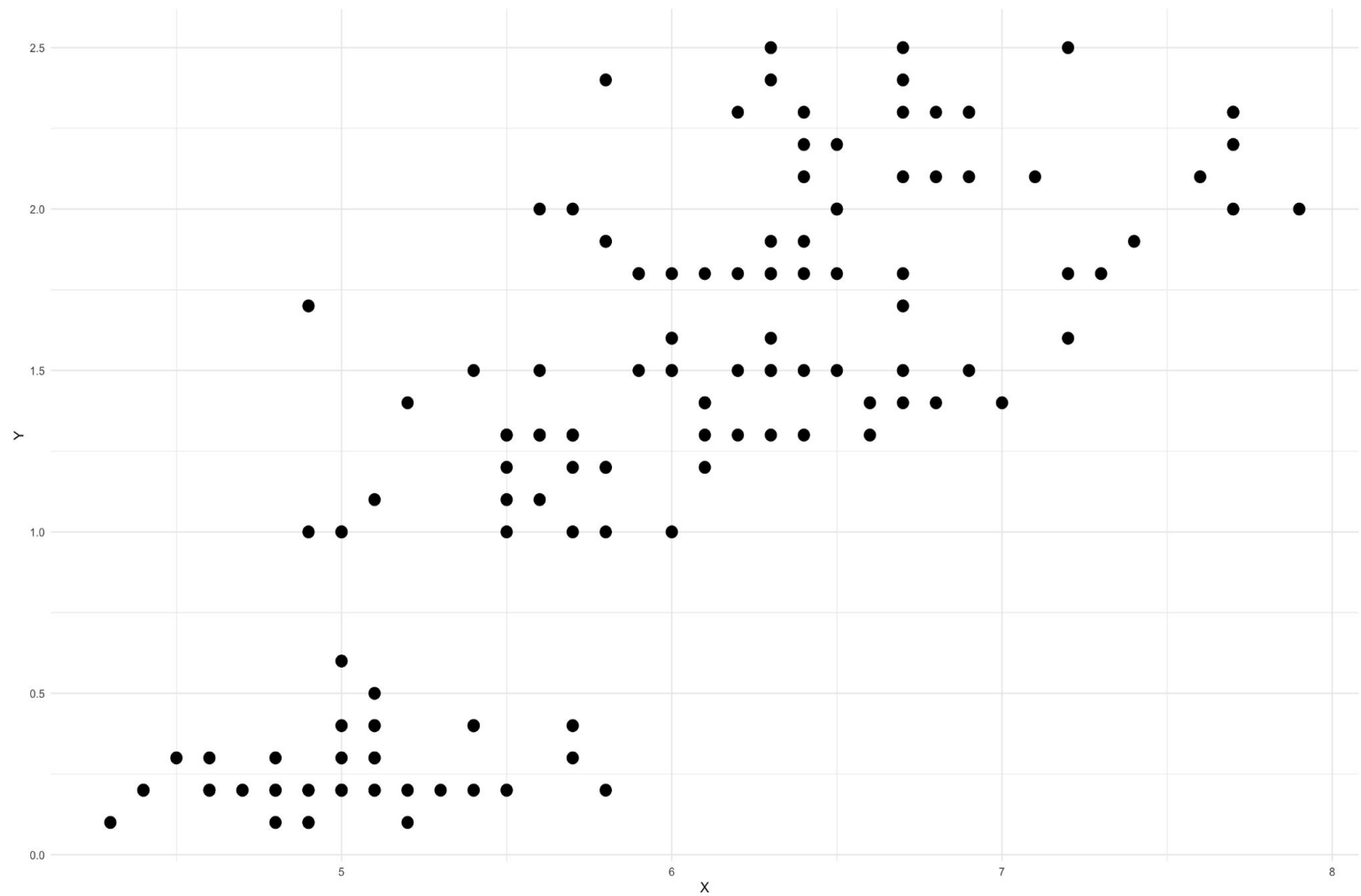
Step 4: Recalculate centroids as the mean of assigned points

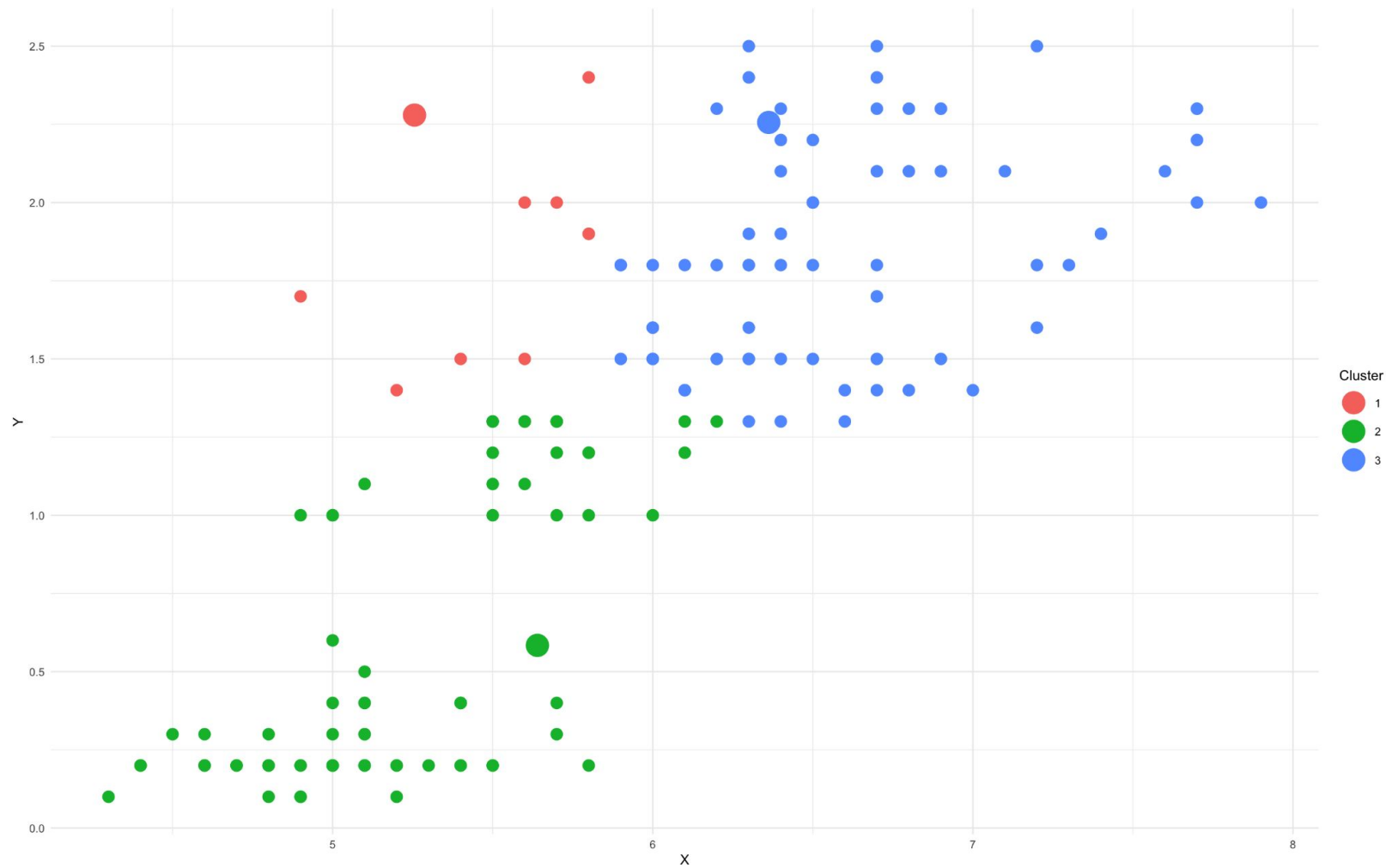
Step 5: Repeat steps 3-4 until convergence

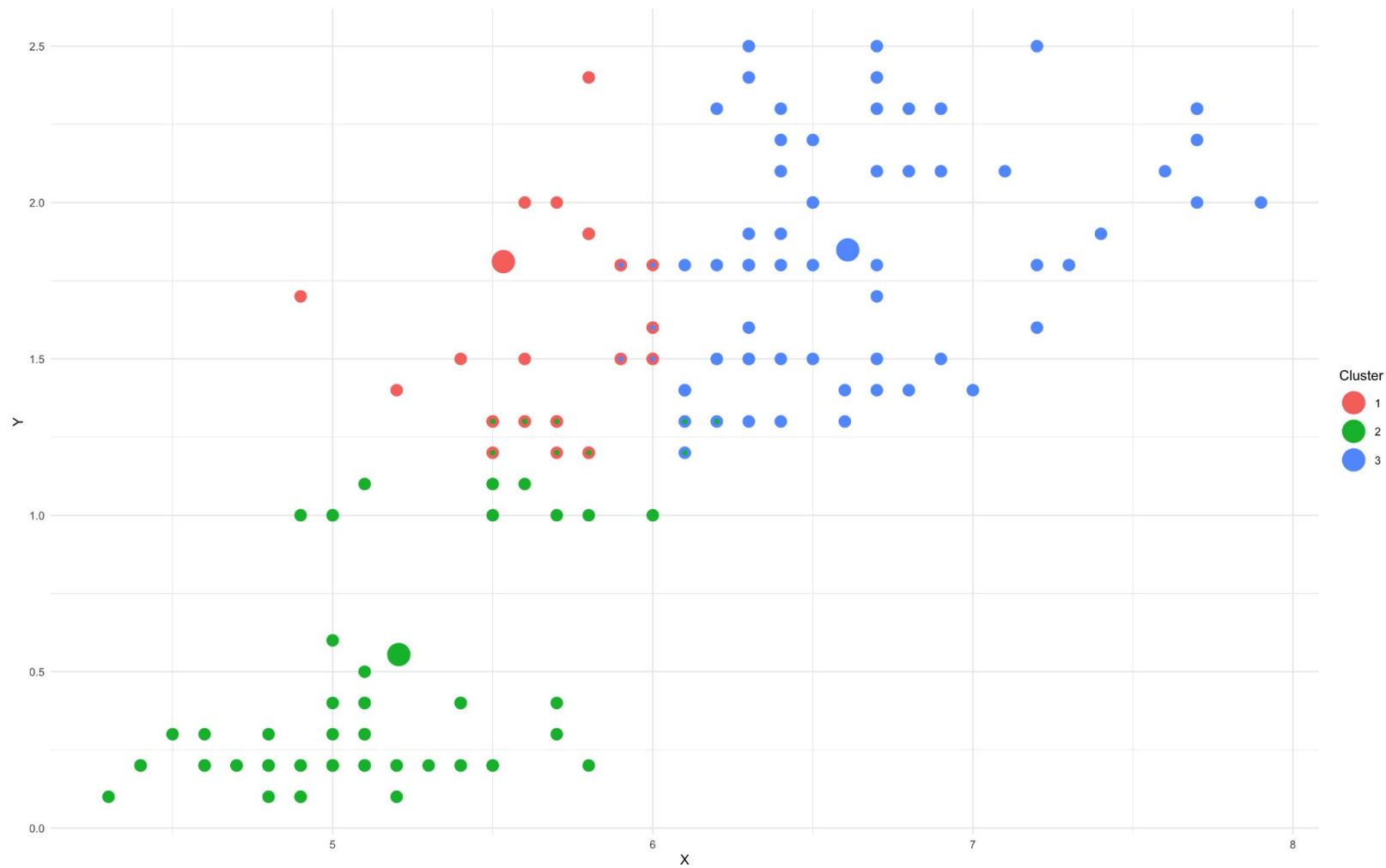
Clusters in iris dataset

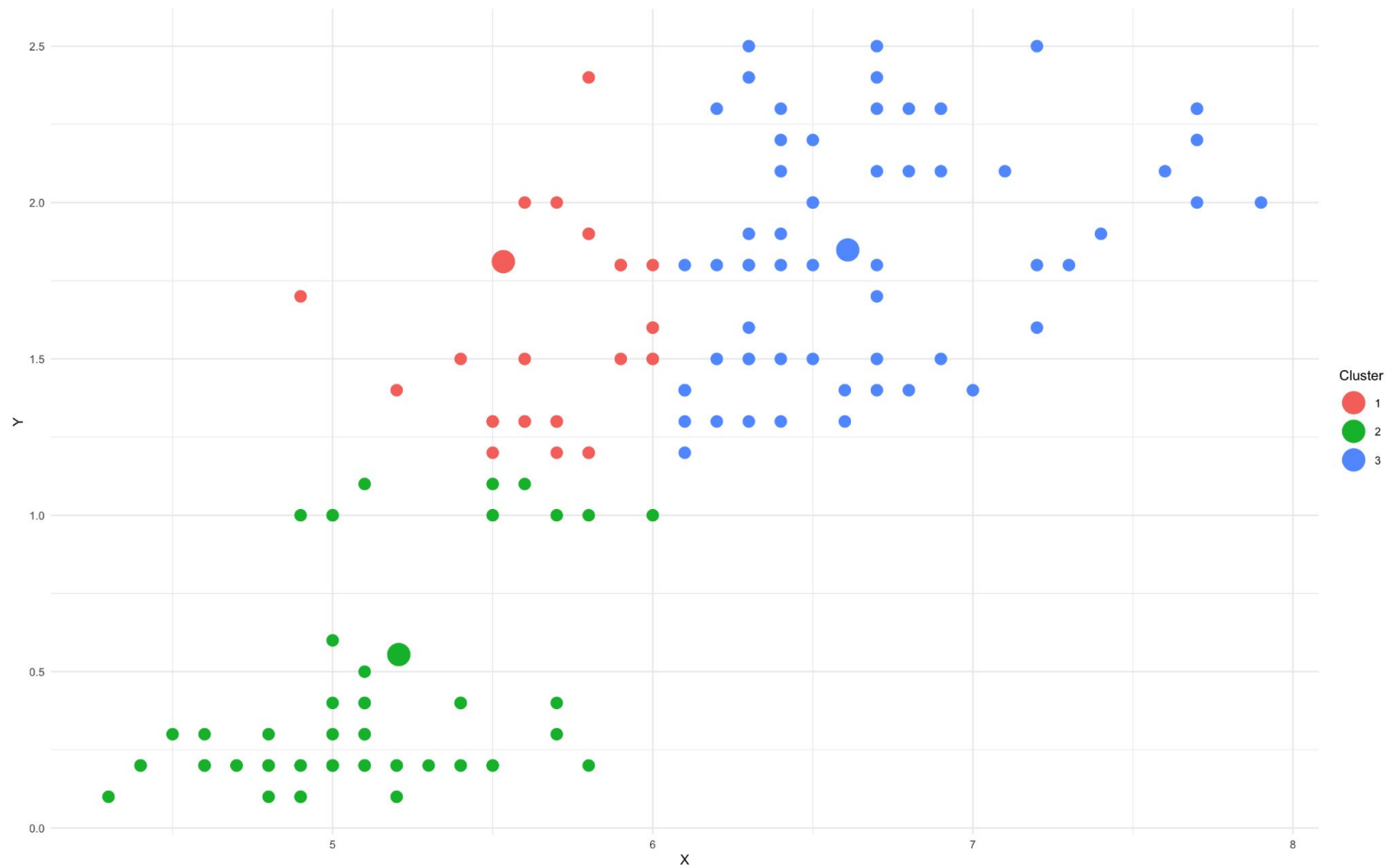
- Let's see if there are clusters in sepal length against petal width from iris dataset
- Let $k = 3$ (there are actually 3 clusters)

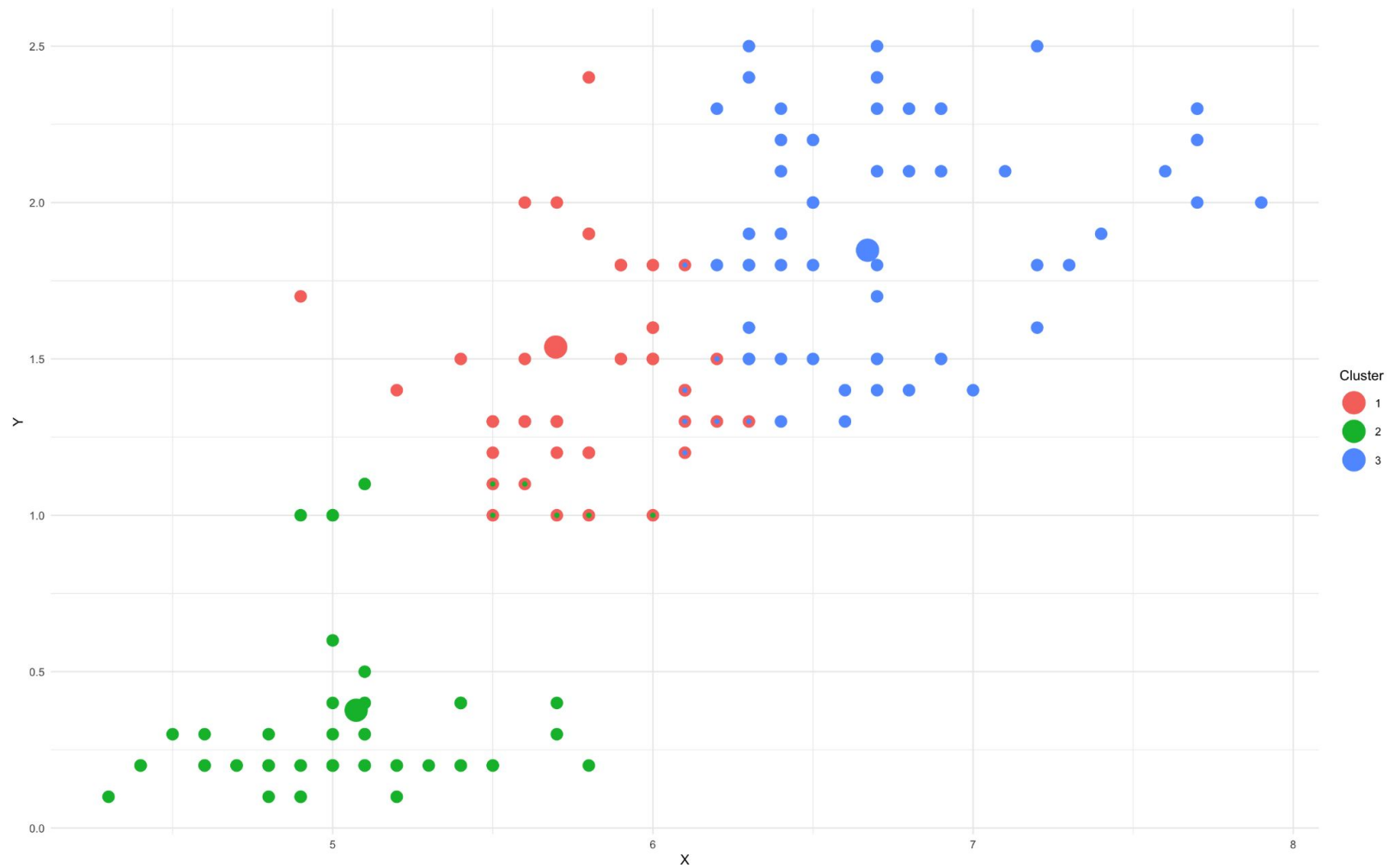
```
set.seed(1)
k_means_visual(k = 3,
               data = iris[,c("Sepal.Length", "Petal.Width")],
               print_plot = TRUE)
```

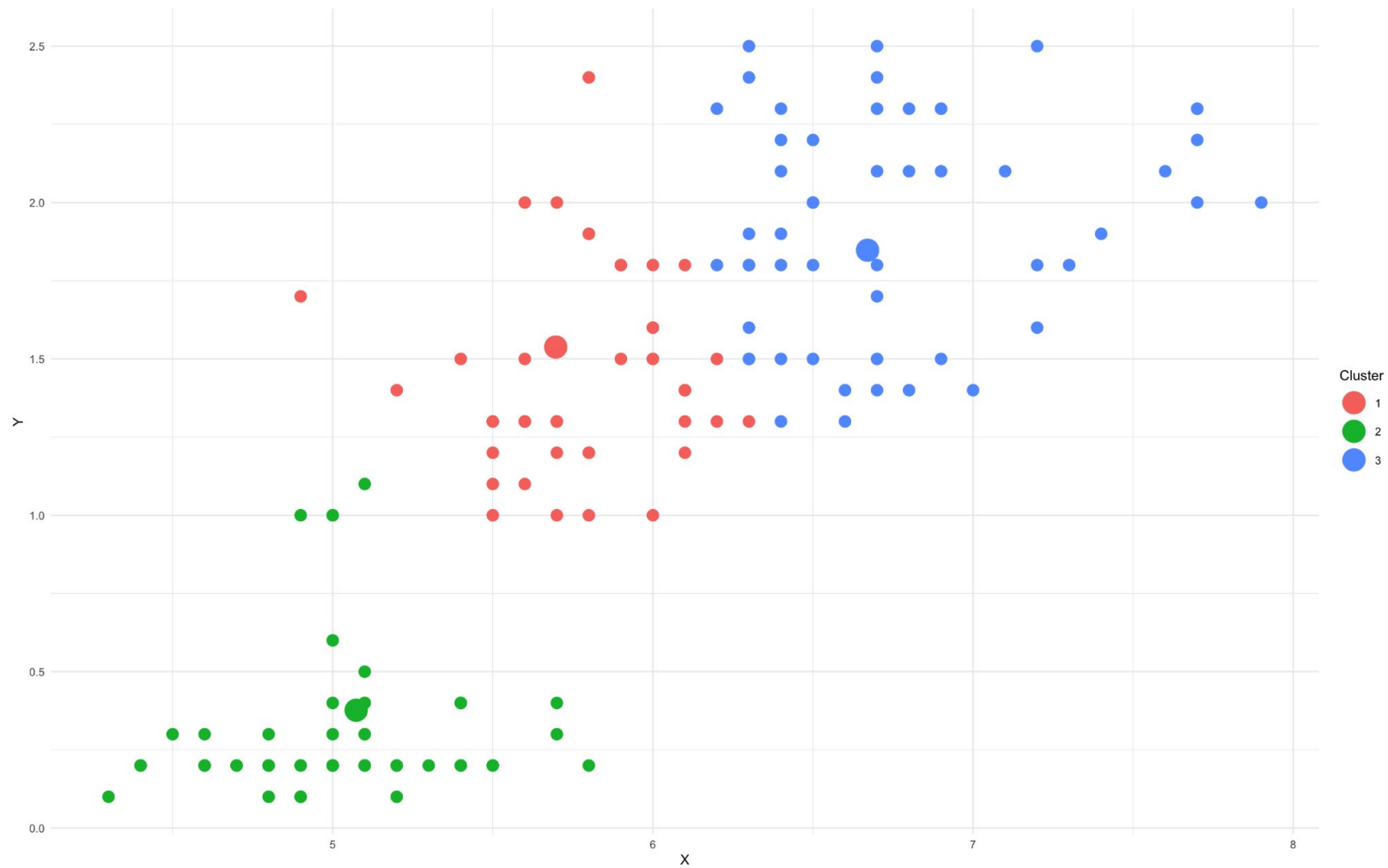


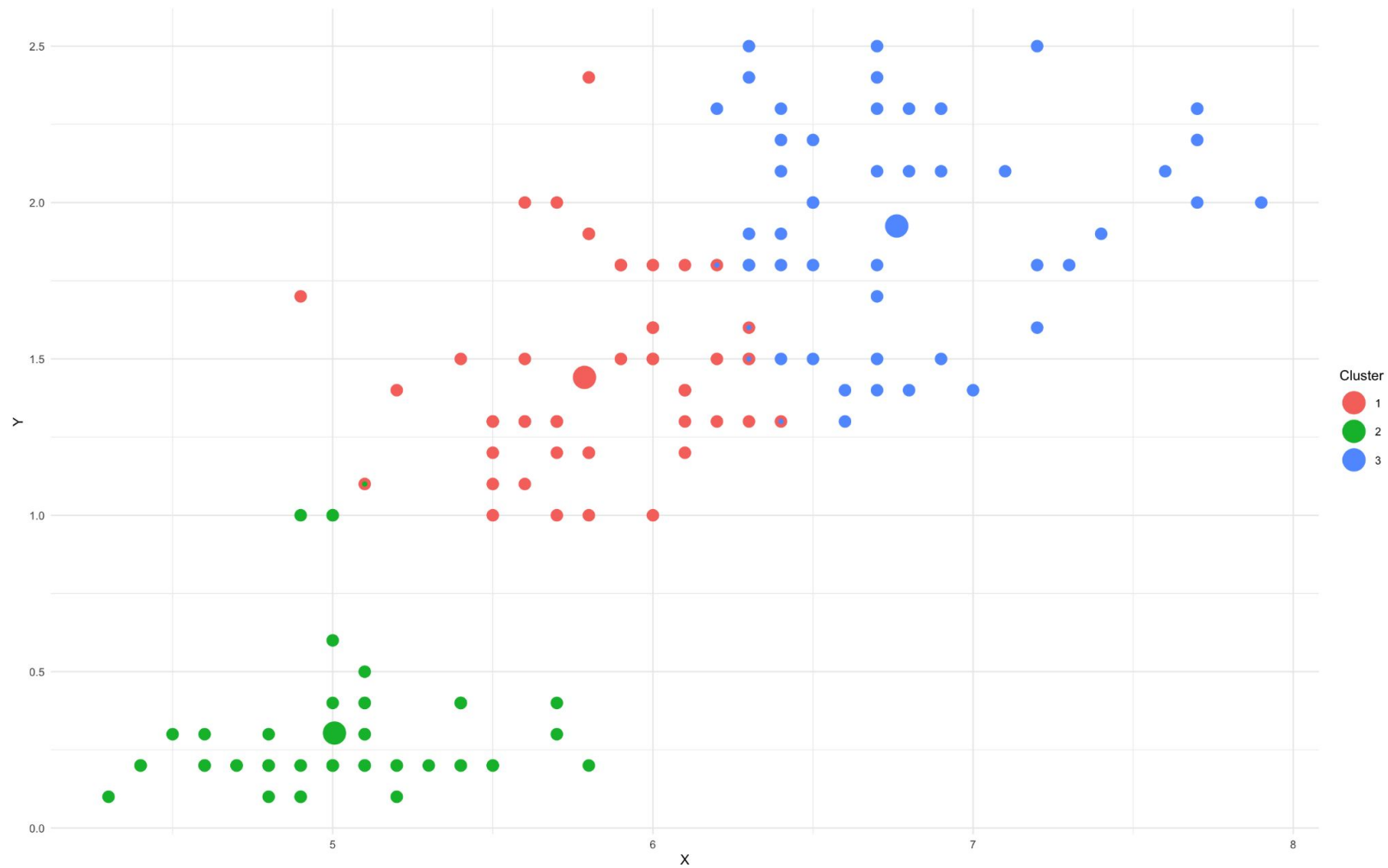


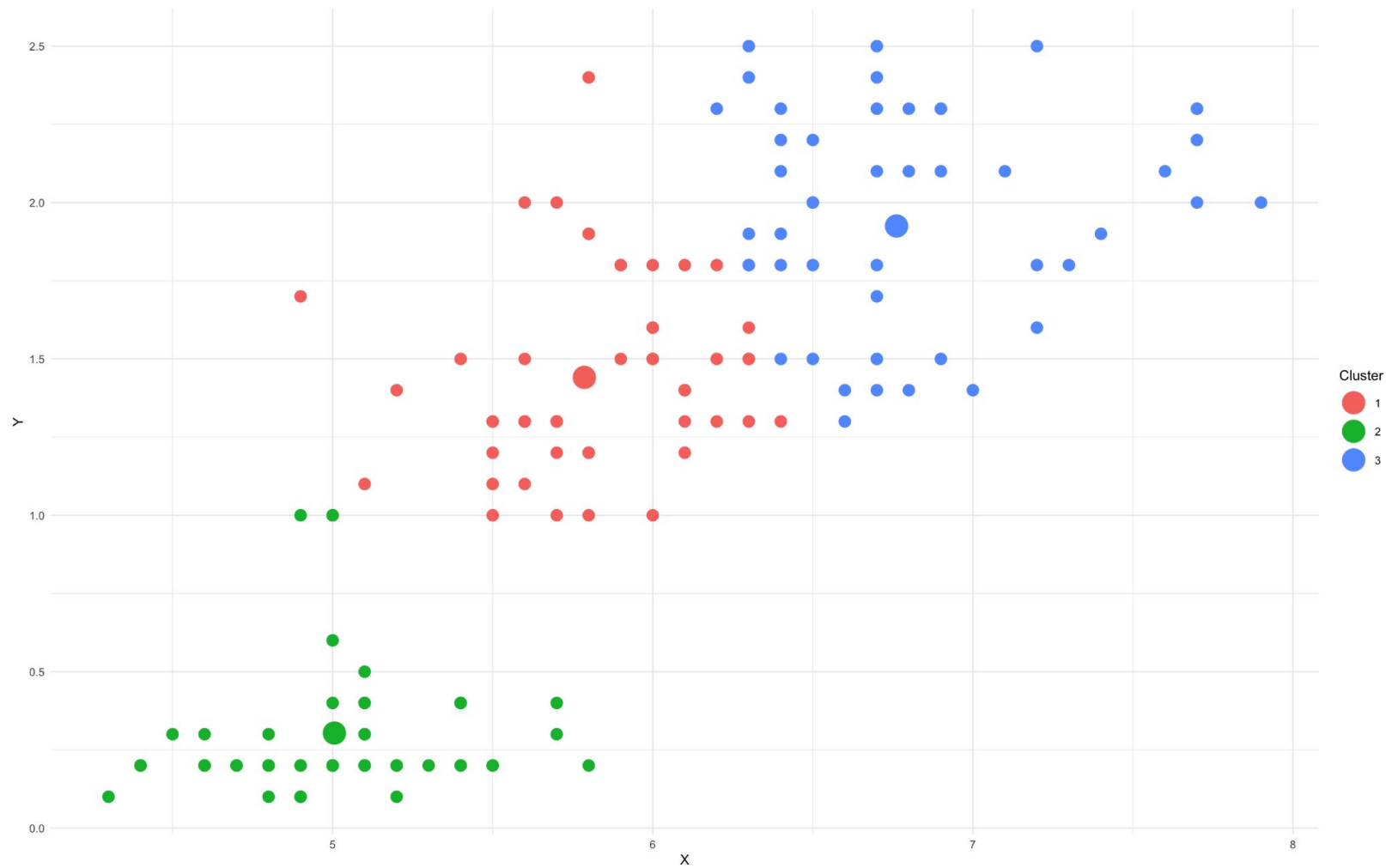


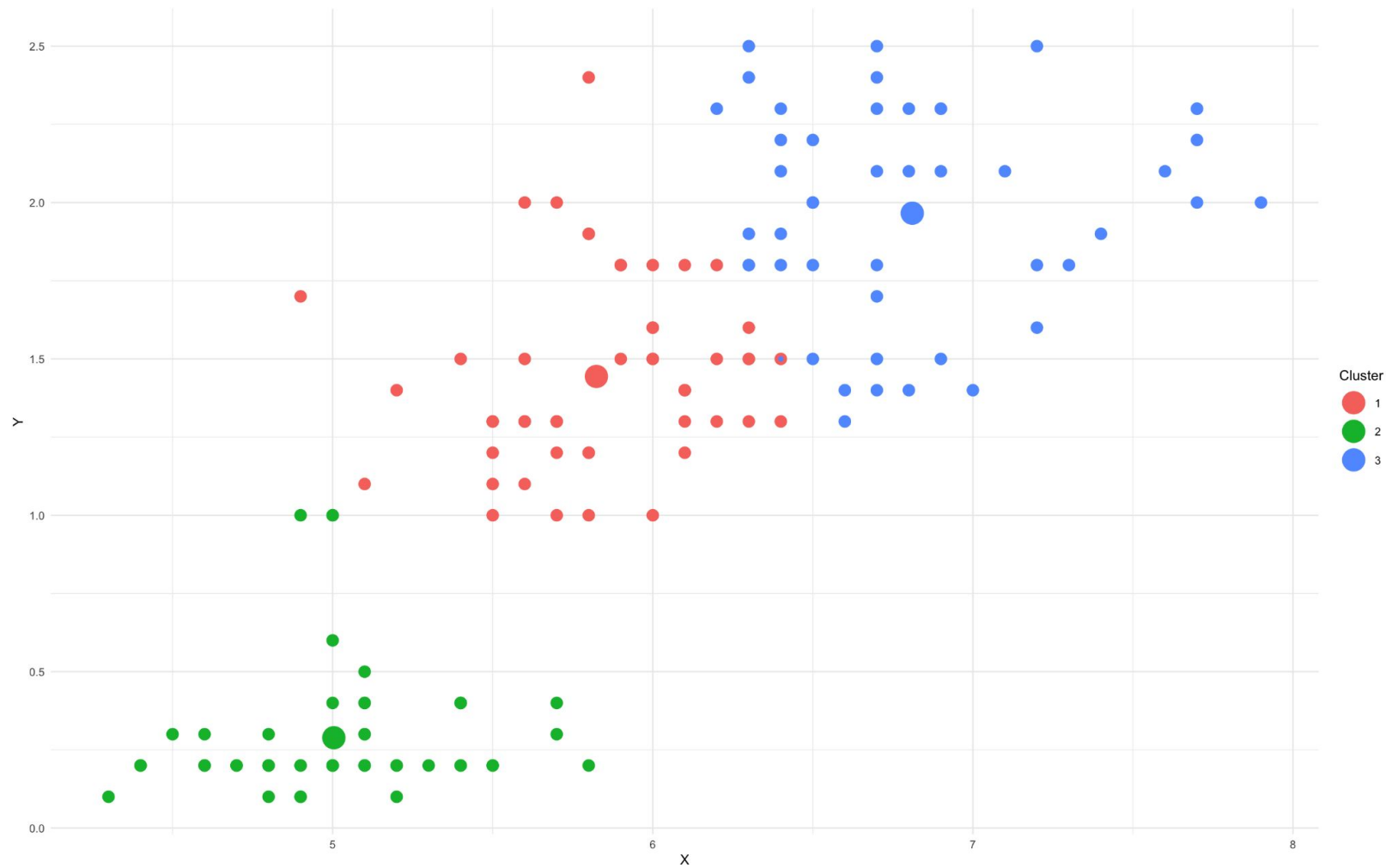


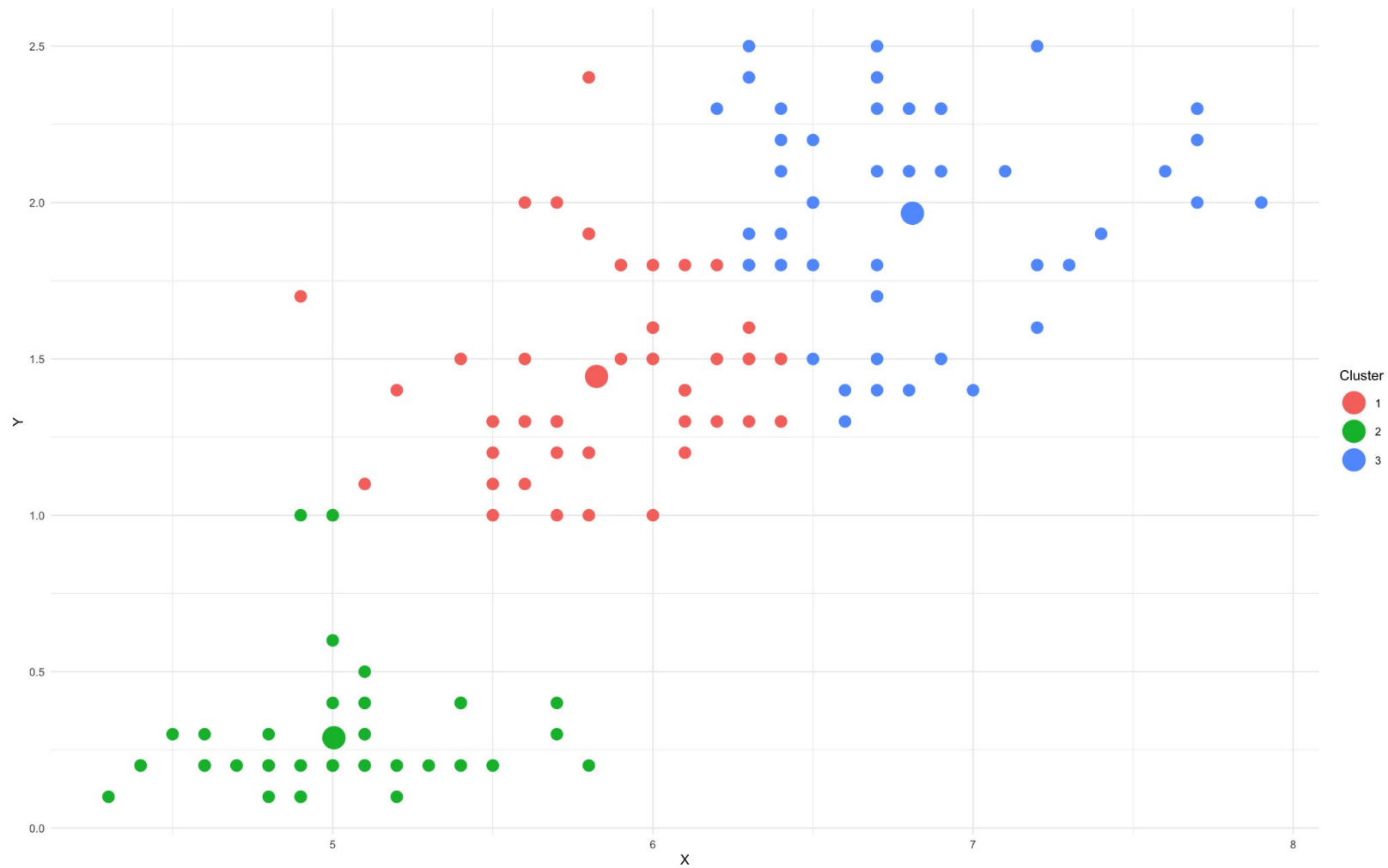


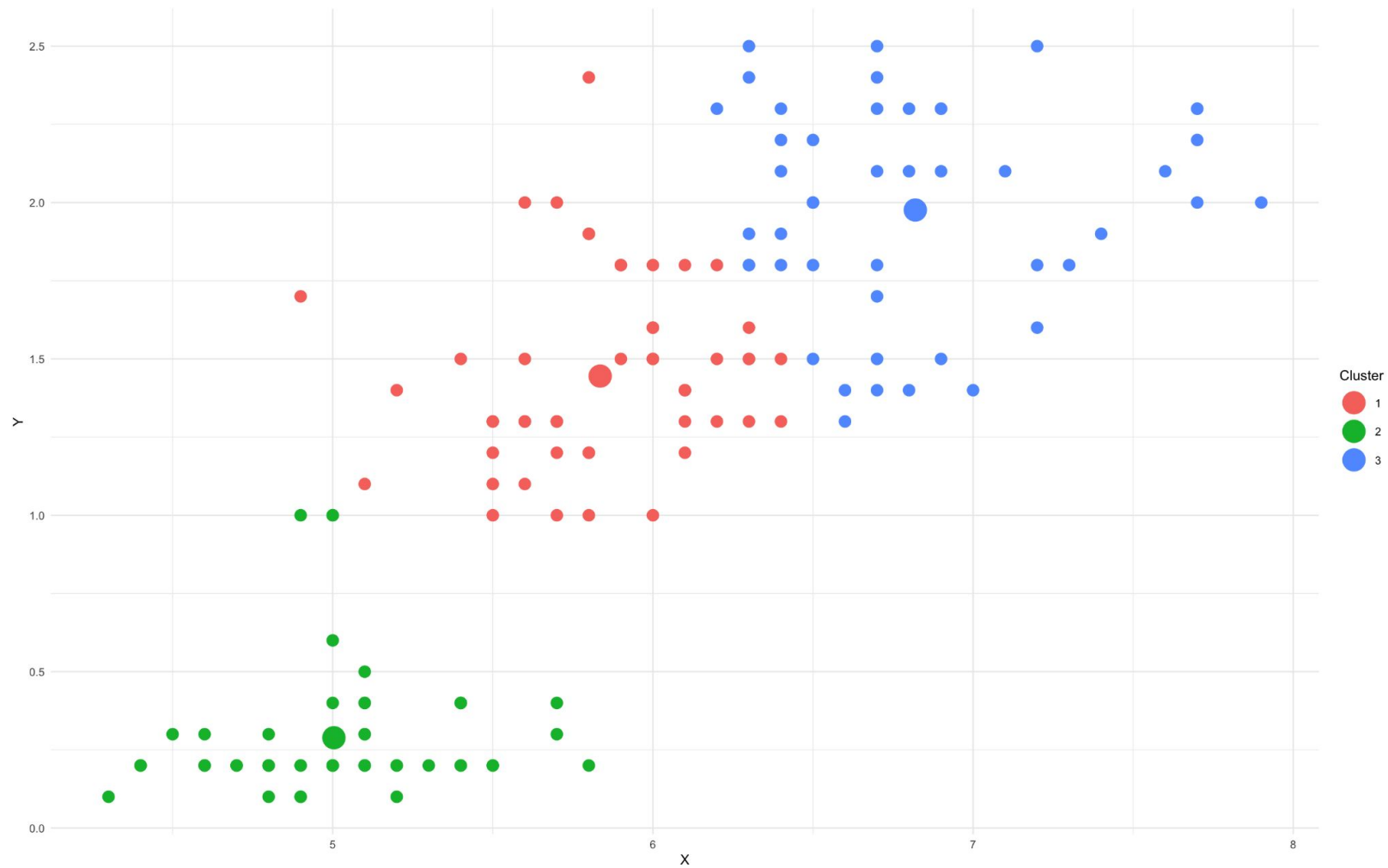


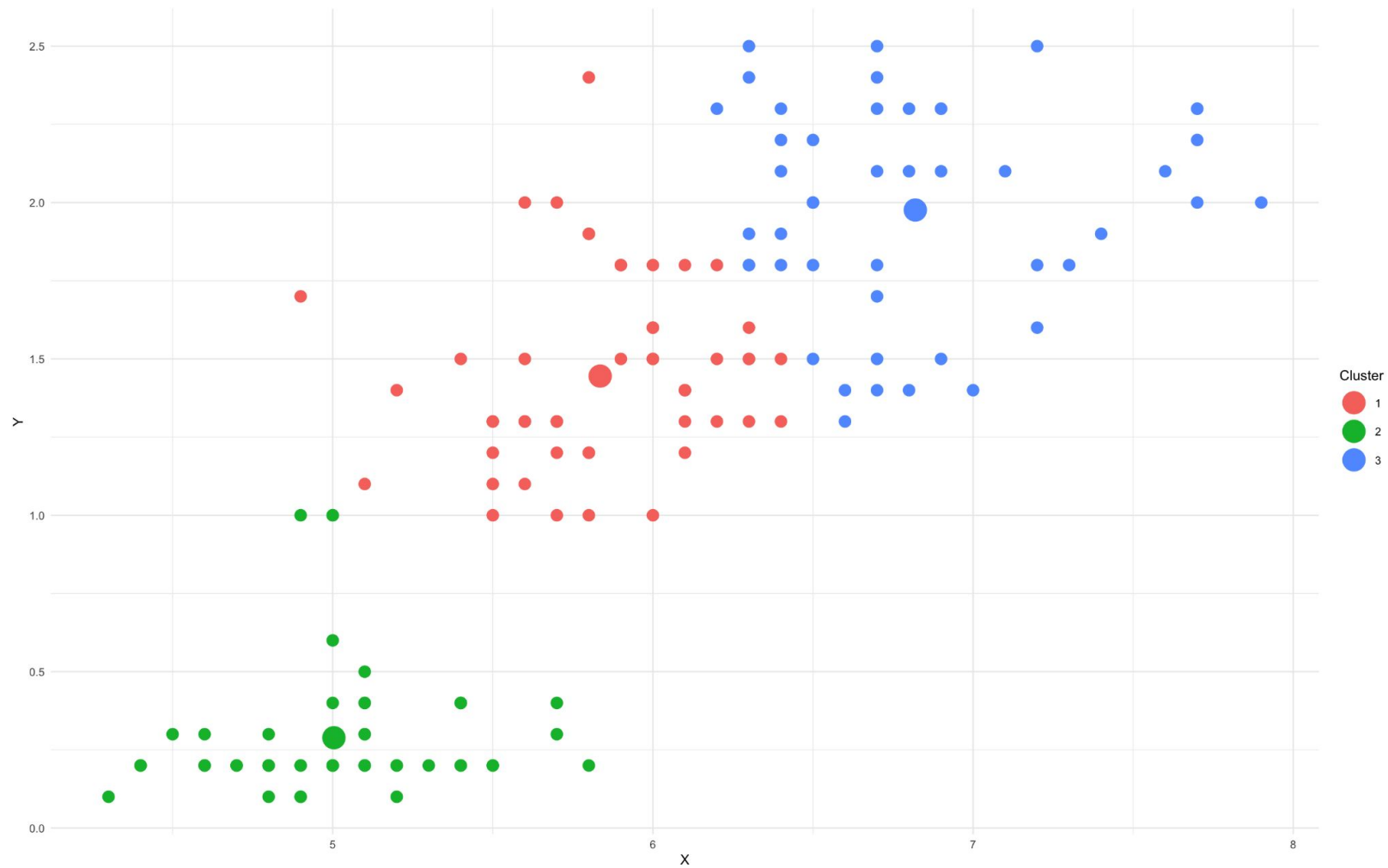












Output

\$clusters

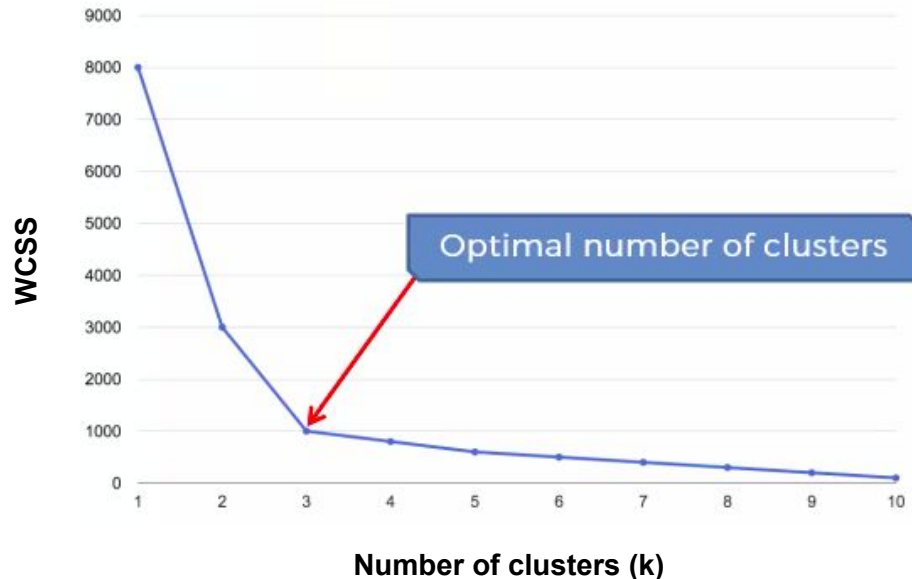
[illegible]

```
$cluster_locations
```

	x	y
[1,]	5.835294	1.4450980
[2,]	5.003774	0.2886792
[3,]	6.819565	1.9760870

How do we choose the best k?

Elbow Method: plot Within-Cluster Sum of Squares (WCSS) vs k and look for the "elbow" point where the rate of decrease sharply shifts



Best k for iris

- Let's now find the best k value using total within-cluster sum of squares
- Let's test k's between 1 and 10

```
set.seed(1)
optimal_k(max_k = 10,
          data = iris[,c("Sepal.Length", "Petal.Width")])
```

